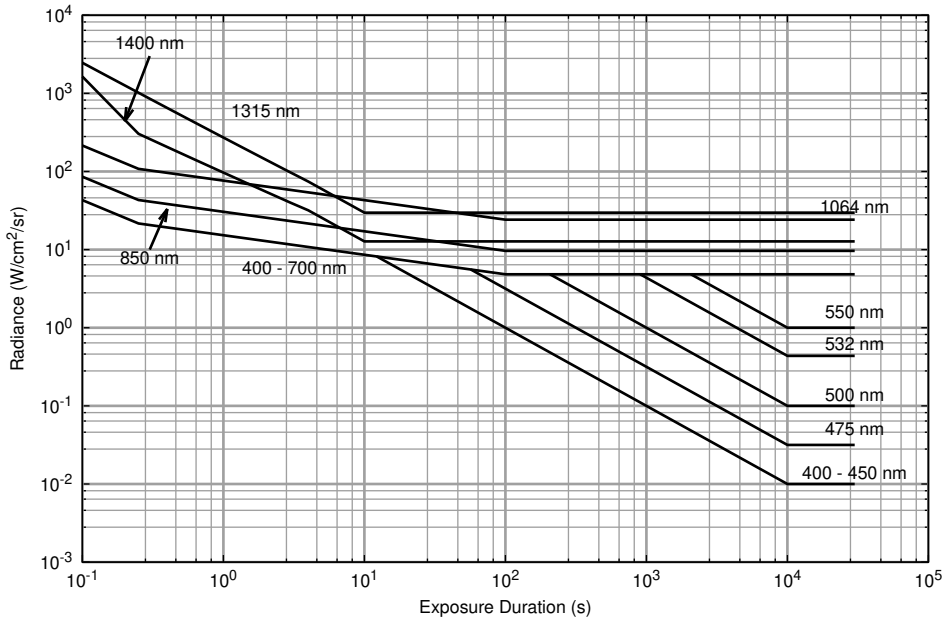
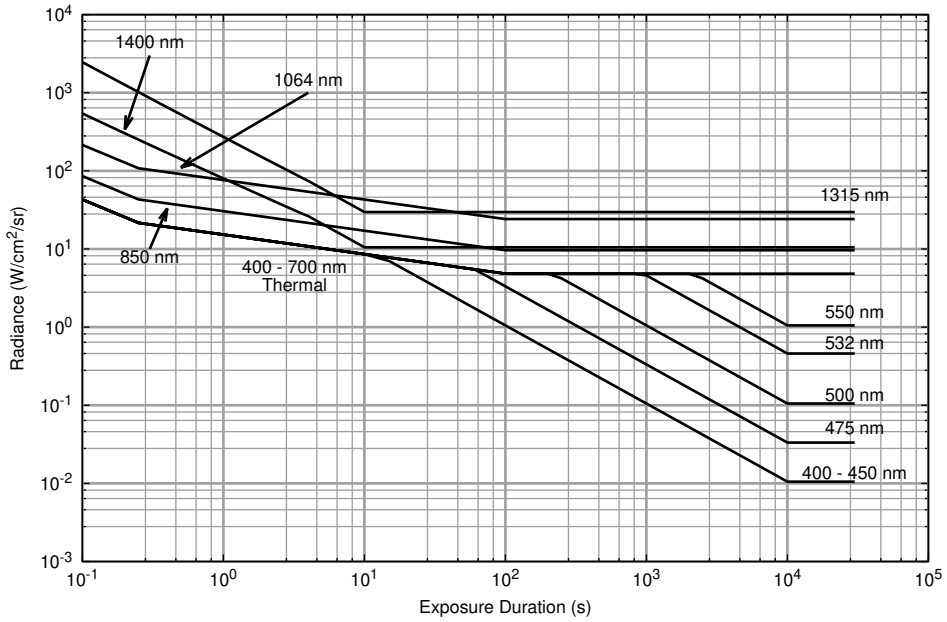
The Problem

In the 2022 version of the standard, Figure 10f looks like this:



However, after we updated the figure's script to calculate the MPE using Python code, it looks like this:



There are two main differences, both with the 1400 nm line. First, the current version does not have a bend between 0.2 and 0.3 seconds. Second, the 1400 nm and 850 nm lines are on top of each other for long times.

In this write up we work through the calculation to see which figure is correct.

The Analysis

The MPE for 700 nm to 1050 nm is

$1 \cdot 10^{-13}$ s to $1 \cdot 10^{-11}$ s	$1.0 C_E \times 10^{-7}$ J cm^{-2}
$1 \cdot 10^{-11}$ s to $5 \cdot 10^{-6}$ s	$2.0 C_A C_E \times 10^{-7}$ J cm^{-2}
$5 \cdot 10^{-6}$ s to T_2	$1.8 C_A C_E t^{0.75} \times 10^{-3}$ J cm^{-2}
T_2 to 30000 s	$1.8 C_A C_E T_2^{-0.25} \times 10^{-3}$ W cm^{-2}

For 1200 nm to 1400 nm, there are dual limits. For the retina the MPE is

$1 \cdot 10^{-13}$ s to $1 \cdot 10^{-11}$ s	$1.0 C_C C_E \times 10^{-7}$ J cm^{-2}
$1 \cdot 10^{-11}$ s to $13 \cdot 10^{-6}$ s	$2.0 C_C C_E \times 10^{-6}$ J cm^{-2}
$13 \cdot 10^{-6}$ s to T_2	$9.0 C_C C_E t^{0.75} \times 10^{-3}$ J cm^{-2}
T_2 to 30000 s	$9.0 C_C C_E T_2^{-0.25} \times 10^{-3}$ W cm^{-2}

For the cornea, the MPE is

$1 \cdot 10^{-11}$ s to $1 \cdot 10^{-3}$ s	$0.3 K_\lambda$ J cm^{-2}
$1 \cdot 10^{-3}$ s to 4 s	$0.3 K_\lambda + 0.56 t^{0.25} - 0.1$ J cm^{-2}
4 s to 10 s	$0.3 K_\lambda + 0.7$ J cm^{-2}
10 s to 30000 s	$0.03 K_\lambda + 0.07$ W cm^{-2}

Let's check the long exposure MPEs first. For 850 nm, MPE : $E = 1.8 C_A C_E T_2^{-0.25} \times 10^{-3}$ W cm^{-2}. The C_A correction factor is $10^{0.002(\lambda - 700)}$ for 850 nm. The C_E correction factor depends on the source size. The figure caption says that this is the MPE for source sizes greater than or equal to 100 mrad, so the correction factor is $\frac{\alpha^2}{\alpha_{\min} \alpha_{\max}}$ with $\alpha_{\min} = 1.5$ mrad and $\alpha_{\max} = 100$ mrad for exposures longer than 0.25 s. The T_2 parameter is 100 s sources larger than 100 mrad

For $\alpha = 100$ mrad then

$C_E = \frac{100^2}{1.5 \times 100} = 66.66666666666667$

$C_A = 10^{0.3} = 1.9952623149688795$

$T_2 = 100^{-0.25} = 0.31622776601683794$

and the MPE is

MPE : $E = 0.0757148813376232$ W cm^{-2}

To convert this to radiance we divide by the irradiance by the solid angle of the source $\Omega = \pi \frac{\alpha^2}{4} = 0.007853981633974483$ sr, so MPE : $L = \frac{0.0757148813376232}{0.007853981633974483} = 9.640318104399222$ W cm^{-1} sr^{-1}. This looks to be where the 850 nm line is.

For 1400 nm we would actually use a different table, but we can pretend that it is in the same table. The C_C correction factor is

$C_C = 8 + 10^{0.04(\lambda - 1250)} = 1000008.0$ (Note that this will change with the next version).

The retina MPE is

MPE : $E = 189738.17750337964$ W cm^{-2}

which, expressed as a radiance is.

MPE : $L = 24158215.07432826$ W cm^{-2} sr^{-1}

This is much larger than the 850 nm line, but we still need to compute the corneal MPE.

$K_\lambda = 10^{0.01(1400-\lambda)} = 10^{0.0} = 1.0$

So the MPE is

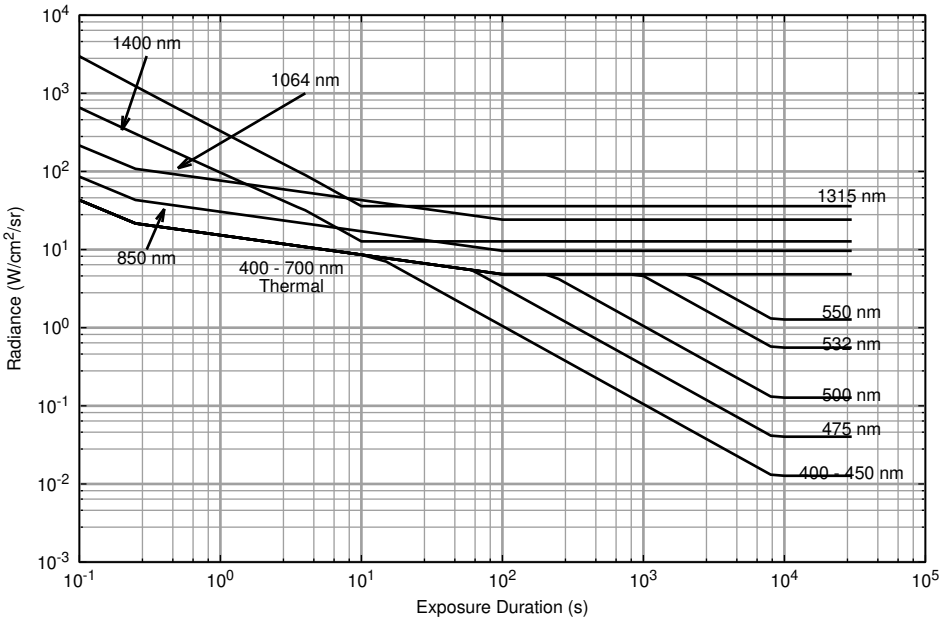
$\text{MPE} : E = 0.1$

and

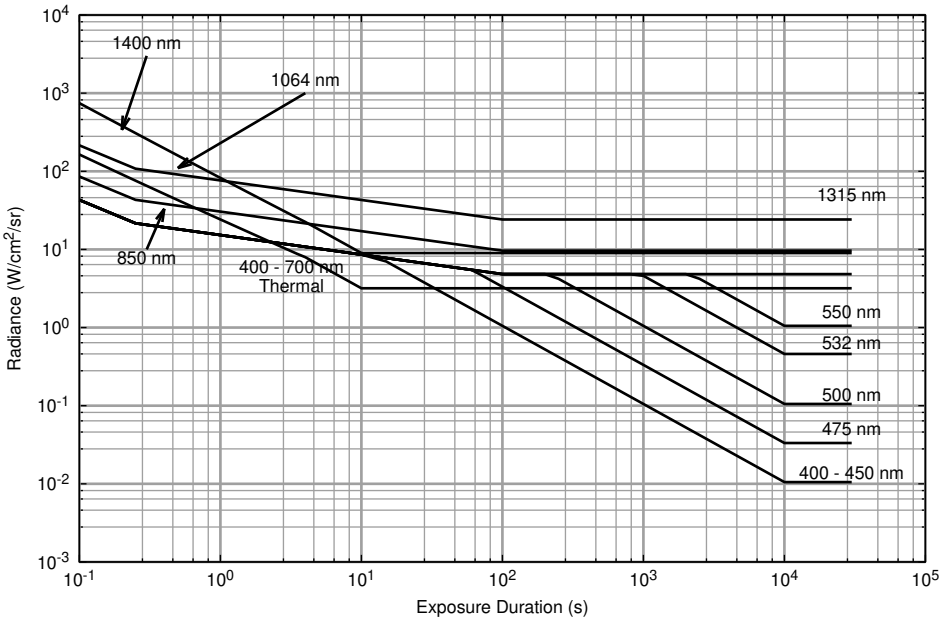
$\text{MPE} : L = 12.732395447351626$

which definitely looks more like the 2022 version. There must be a bug in the code…

…It turns out that the script generating the data for the figure was using an apparent size of 110 mrad. If we change the source size to 100 mrad we get the following figure



This makes the 1400 nm line match (except for the bend on the left), but the MPEs for long exposure times are higher. If we change the source size to 200 mrad we get this figure



There the 1315 nm and 1400 nm lines have both been shifted down.

It is likely that this has to do with the application of a limiting cone angle that was necessary in previous versions of the standard but are no longer needed since the introduction of C_g. To convert the MPE to radiance, we divide by the solid angle of the source.

It turns out that this has to do with the cornea dual limits. In the 2007 version of the standard, there were no cornea limits in the NIR, so this figure just showed retinal limits. The NIR limits were all still given in terms of radiant exposure/irradiance (the large source photochemical limits in the visible were given in terms of integrated radiance/radiance for the short wavelengths). To convert from radiant exposure/irradiance to integrated radiance/radiance, we divide by the solid angle of the source

$$\text{MPE} : L = \text{MPE} : \frac{E}{\Omega} = 4\text{MPE} : \frac{E}{\pi\alpha^2}$$

For extended source, the retinal MPE's all contain the C_E correction factor, which, for sources larger than 100 is $C_E = \frac{\alpha^2}{1.5\times 100}$. So MPE as an integrated radiance/radiance will be

$$\text{MPE} : L = C_E \frac{A(t)}{\Omega} = \frac{\alpha^2}{1.5 \times 100} \times A(t) \times \frac{4}{\pi\alpha^2} = \frac{4A(t)}{\pi \times 1.5 \times 100}$$

where $A(t)$ is some function of time that depends on the wavelength. This shows that the MPE, expressed as integrated radiance/radiance, does not depend on the source size for large sources. which is the point of expressing them that way.

The Solution

The value is questionable. The original version was basically showing that the MPEs became a constant integrated radiance/radiance with source size after a certain point (α_{max}). There are three different options for modifying the figure to make it "correct".

1. Remove the 1200 nm to 1400 nm wavelengths.
2. Only display the retinal MPEs. The figure caption would need to be updated to indicate this.
3. Pick a source size and change the figure caption from ">100 mrad" to indicate the source size used.
4. Use γ to convert to integrated radiance/radiant exposure.

Based on the original intent, option seems to make the most sense. With option 2 we would be displaying limits that don't actually apply for the longer wavelengths because the cornea limits are lower for large sources. With option 3 we lose the value of saying "all sources larger than this are the same". Option 4 would make since in that is similar to using the limiting aperture to the MPE as a power or energy, but it must be interpreted "the MPE averaged over a limiting cone angle of…". Similar to how the MPE expressed as a radiant exposure/irradiance is interpreted as "the MPE averaged over a limiting aperture of…". If we do this though, a user computing the MPE would have to know that for $\alpha > \gamma$ you use γ to compute C_E, not α, which may be confusing since the radiant exposure/irradiance is allowed to increase with source size.